CSEC Mathematics

January 2018 – Paper 2

Solutions

Kerwin Springer

Best Online Lessons in the Caribbean
WhatsApp **+1868-310-1306** for more information

SECTION I

Answer ALL questions in this section.

All working must be clearly shown.

1. (a) Using a calculator, or otherwise, calculate

(i) $5\frac{1}{2} \div 3\frac{2}{3} + 1\frac{4}{5}$, giving your answer as a fraction in its lowest terms. [2]

$$= 5\frac{1}{2} \div 3\frac{2}{3} + 1\frac{4}{5}$$

$$= \left(\frac{11}{2} \div \frac{11}{3}\right) + \frac{9}{5}$$

$$= \left(\frac{11}{2} \times \frac{3}{11}\right) + \frac{9}{5}$$

$$= \frac{3}{2} + \frac{9}{5}$$

$$= \frac{15}{10} + \frac{18}{10}$$

$$= \frac{33}{10}$$ [As a fraction in its lowest terms]

Kerwin Springer

(ii) 165×0.38^2 , giving your answer as an EXACT value. [1]

$165 \times 0.38^2 = 165 \times 0.38 \times 0.38$

$= 23.826$ [In exact form]

(b) Write your answer in (a)(ii) correct to

(i) two decimal places [1]

$23.826 \approx 23.83$ [to two decimal places]

(ii) three significant figures [1]

$23.826 \approx 23.8$ [to three significant figures]

(iii) the nearest whole number. [1]

$23.826 \approx 24$ [to the nearest whole number]

Kerwin Springer

(c) Mr. Adams invested \$5 000 at the credit union and received \$5 810, inclusive of simple interest, after 3 years.

Determine

(i) the simple interest earned [1]

Simple Interest earned = Amount received − Principal amount

= \$5 810 − \$5 000

= \$810

(ii) the annual interest rate paid by the credit union [2]

Substituting Principal = \$5000, Simple Interest = \$810, Time = 3 years into:

$$SI = \frac{P \times R \times T}{100}$$

$$810 = \frac{5000 \times R \times 3}{100}$$

$$810 = 150R$$

$$\frac{810}{150} = R$$

$$R = \frac{810}{150}$$

$$R = 5.4\%$$

∴ The annual interest paid by the credit union is 5.4%.

Kerwin Springer

(iii) the length of time it will take for Mr. Adam's investment to be **doubled**, at the same rate of interest. [2]

The investment is \$5000. For the investment to be doubled, the interest amount will have to be \$5000.

$$SI = \frac{P \times R \times T}{100}$$

$$5000 = \frac{5000 \times 5.4 \times T}{100}$$

$$5000 = 270\,T$$

$$\frac{5000}{270} = T$$

$$T = \frac{5000}{270}$$

$T = 18.52$ years (correct to 2 decimal places)

Total: 11 marks

Kerwin Springer

2. (a) Given that $a * b$ means $\sqrt{a + 4b}$, where the positive root is taken, determine

(i) the value of $1 * 2$ [2]

$a * b = \sqrt{a + 4b}$

$= \sqrt{1 + 4(2)}$

$= \sqrt{1 + 8}$

$= \sqrt{9}$

$= 3$ [taking the positive root]

(ii) whether the operation denoted by $*$ is commutative. Justify your answer.

[3]

Let $a = 2$ and $b = 3$,

$a * b = \sqrt{a + 4b}$

$= \sqrt{(2) + 4(3)}$

$= \sqrt{2 + 12}$

$= \sqrt{14}$

$= \sqrt{14}$

$b * a = \sqrt{b + 4a}$

$= \sqrt{(3) + 4(2)}$

$= \sqrt{3 + 8}$

$= \sqrt{11}$

$= \sqrt{11}$

Since $a * b \neq b * a$, then the operation $*$ is not commutative.

Kerwin Springer

(b) (i) Solve the inequality $3 - 2x > 5$. [2]

$3 - 2x > 5$

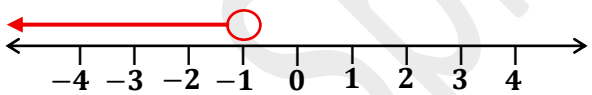
$-2x > 5 - 3$

$-\ \ 2x > 2$

$x < \frac{2}{-2}$

$x < -1$

(ii) Represent your answer in (b)(i) on the number line shown below. [1]



(c) **Statement one:** Two adult tickets and three children tickets cost \$43.00.

Statement two: One adult ticket and one ticket for a child cost \$18.50.

(i) Let x represent the cost of an adult ticket and y the cost of a ticket for a child. Write TWO equations in x and y to represent the information above. [2]

Let x represent the cost of an adult ticket.

Let y represent the cost of a child ticket.

Kerwin Springer

Since 2 adult tickets at $\$x$ each and 3 children tickets at $\$y$ each cost $43.00, then the equation is: $2x + 3y = 43$

Since 1 adult ticket at $\$x$ each and 1 child ticket at $\$y$ each cost $18.50, then the equation is: $x + y = 18.50$

(ii) Solve the equations to determine the cost of an adult ticket. [2]

$2x + 3y = 43$ ...Equation 1

$x + y = 18.50$ ...Equation 2

Rearranging Equation 2 to get Equation 3:

$x = 18.50 - y$ ...Equation 3

Substitute Equation 3 into Equation 1:

$$2(18.50 - y) + 3y = 43$$

$$37 - 2y + 3y = 43$$

$$37 + y = 43$$

$$y = 43 - 37$$

$$y = 6$$

Kerwin Springer

Substitute $y = 6$ into Equation 3:

$x = 18.50 - (6)$

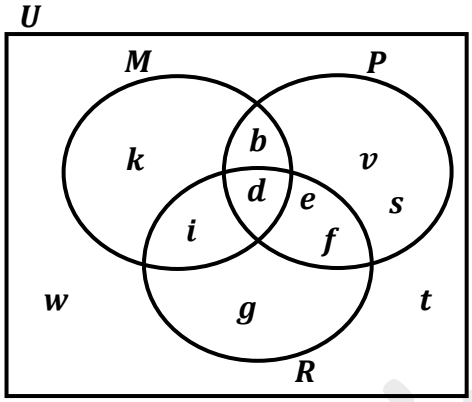
$x = 12.5$

$\therefore$ The cost of 1 adult ticket is \$12.50.

Total: 12 marks

Kerwin Springer

3. (a) The universal set $U = \{b, d, e, f, g, i, k, s, t, v, w\}$. The Venn diagram below shows U and three sets, M, P and R, which are subsets of U.



(i) State the value of $n(P \cup R)$. [1]

$P \cup R = \{b, v, s, d, e, f, i, g\}$

$\therefore n(P \cup R) = 8$

(ii) List the members of

(a) $M \cap P$ [2]

$M \cap P = \{b, d\}$

Kerwin Springer

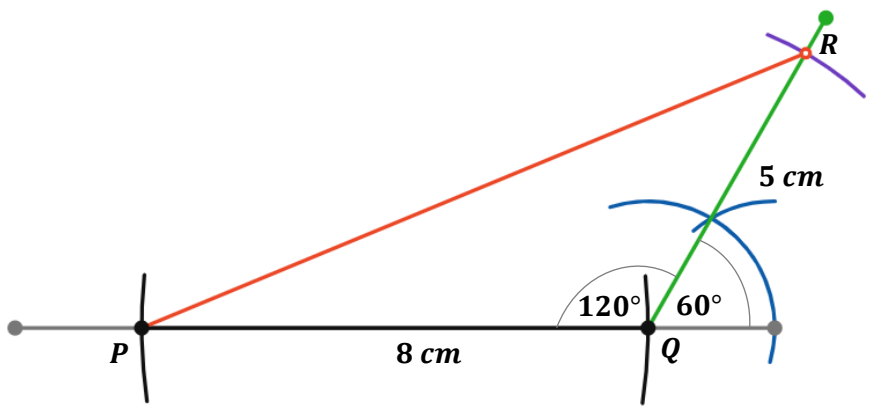
(b) $M \cup R'$ [2]

$M = \{k, b, i, d\}$

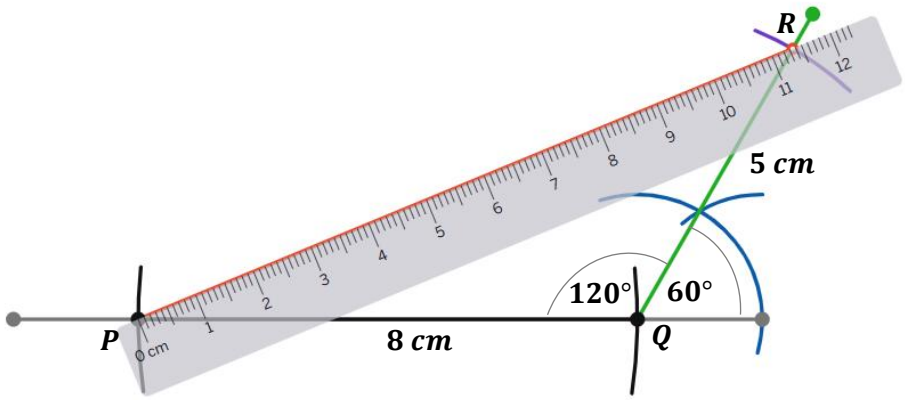
$R' = \{k, b, v, s, t, w\}$

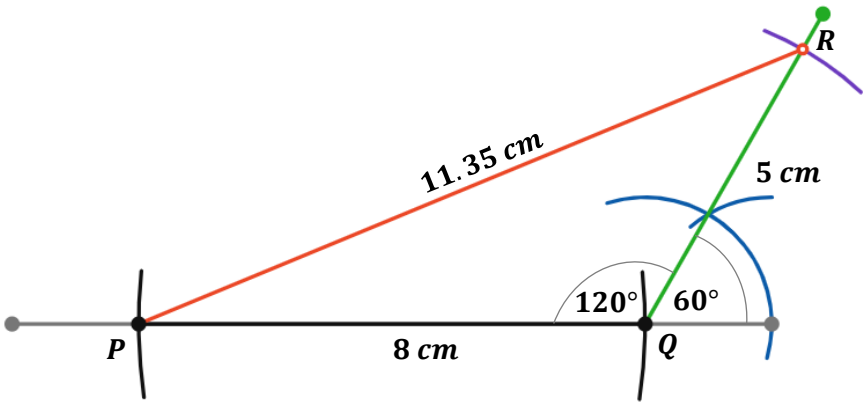
$M \cup R' = \{k, b, i, d, v, s, t, w\}$

(b)(i) Using a ruler, a pencil and a pair of compasses, construct triangle PQR with $PQ = 8$ cm, angle $PQR = 120°$ and $QR = 5$ cm. [4]



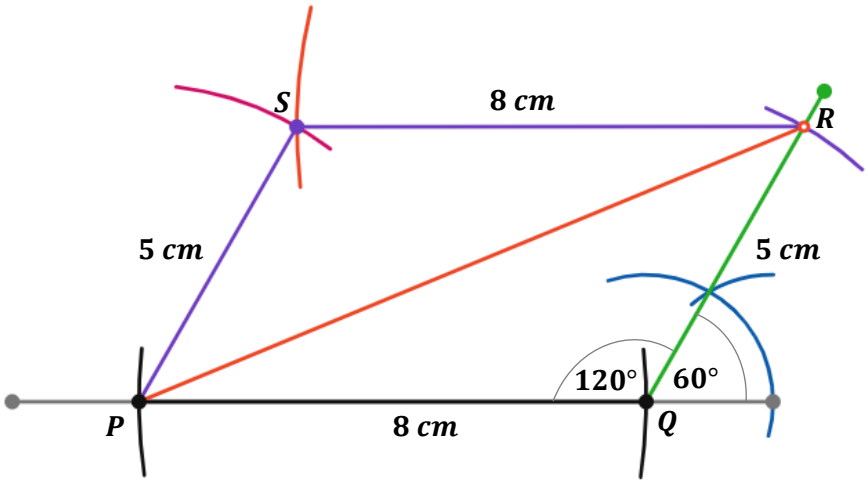
(ii) Measure and state the length of the side PR. [1]





The length of the side $PR = 11.35\ cm$

(iii) On your diagram in (b)(i), construct the point S, such that $PQRS$ forms a parallelogram. [2]



Total: 12 marks

Kerwin Sp

4. (a) The equation of a straight line, l, is given as $3x - 4y = 5$.

(i) Write the equation of the line, l, in the form $y = mx + c$. [2]

$3x - 4y = 5$

$-4y = -3x + 5$

Dividing throughout by -4 gives:

$y = \frac{3}{4}x - \frac{5}{4}$

which is of the form $y = mx + c$, where $m = \frac{3}{4}$ and $c = -\frac{5}{4}$.

(ii) Hence, determine the gradient of the line, l. [1]

$y = \frac{3}{4}x - \frac{5}{4}$

The gradient of the line, l, is $m = \frac{3}{4}$.

(iii) The point P with coordinates $(r, 2)$ lies on the line l. Determine the value of r. [2]

$y = \frac{3}{4}x - \frac{5}{4}$

Substituting point $P(r, 2)$ gives:

$2 = \frac{3}{4}r - \frac{5}{4}$

$8 = 3r - 5$

Kerwin Springer

$3r = 8 + 5$

$3r = 13$

$r = \frac{13}{3}$

(iv) Find the equation of the straight line passing through the point $(6, 0)$ which is perpendicular to l. [2]

The gradient of line l is $m = \frac{3}{4}$.

The gradient of the line perpendicular to $l = \frac{-1}{\frac{3}{4}}$

$= -\frac{4}{3}$

Recall: $y - y_1 = m(x - x_1)$

Substituting $m = -\frac{4}{3}$ and the point $(6, 0)$ gives:

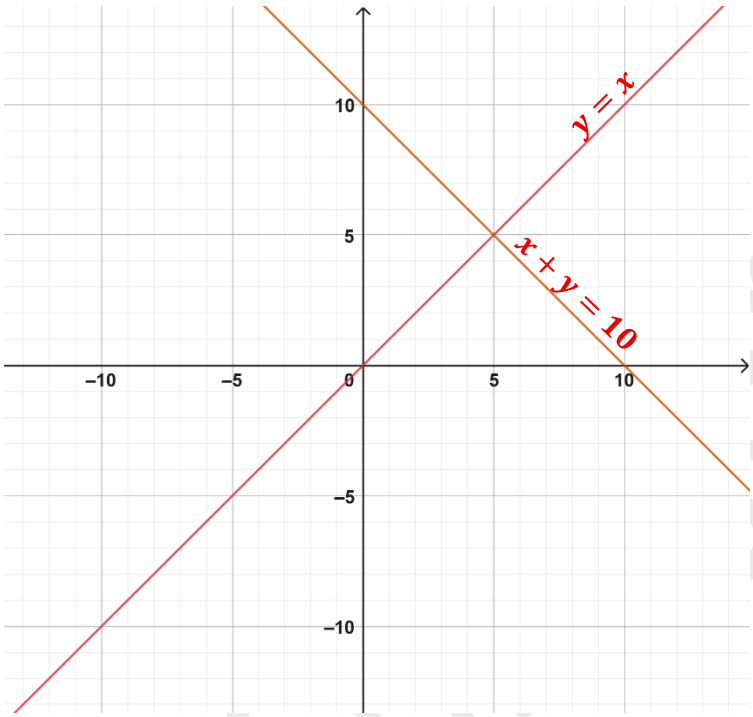
$y - 0 = -\frac{4}{3}(x - 6)$

$y = -\frac{4}{3}x + 8$

$\therefore$ The equation of the required line is $y = -\frac{4}{3}x + 8$.

Kerwin Springer

(b)(i) Draw the straight lines $x + y = 10$ and $y = x$ on the grid below. [2]



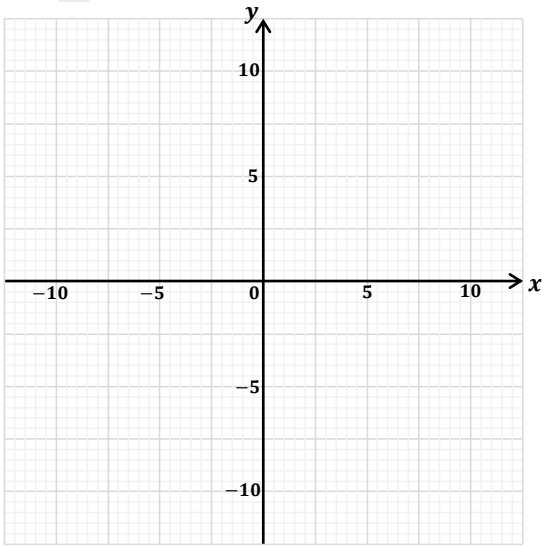
(ii) On the same grid, shade the region which satisfies the FOUR inequalities

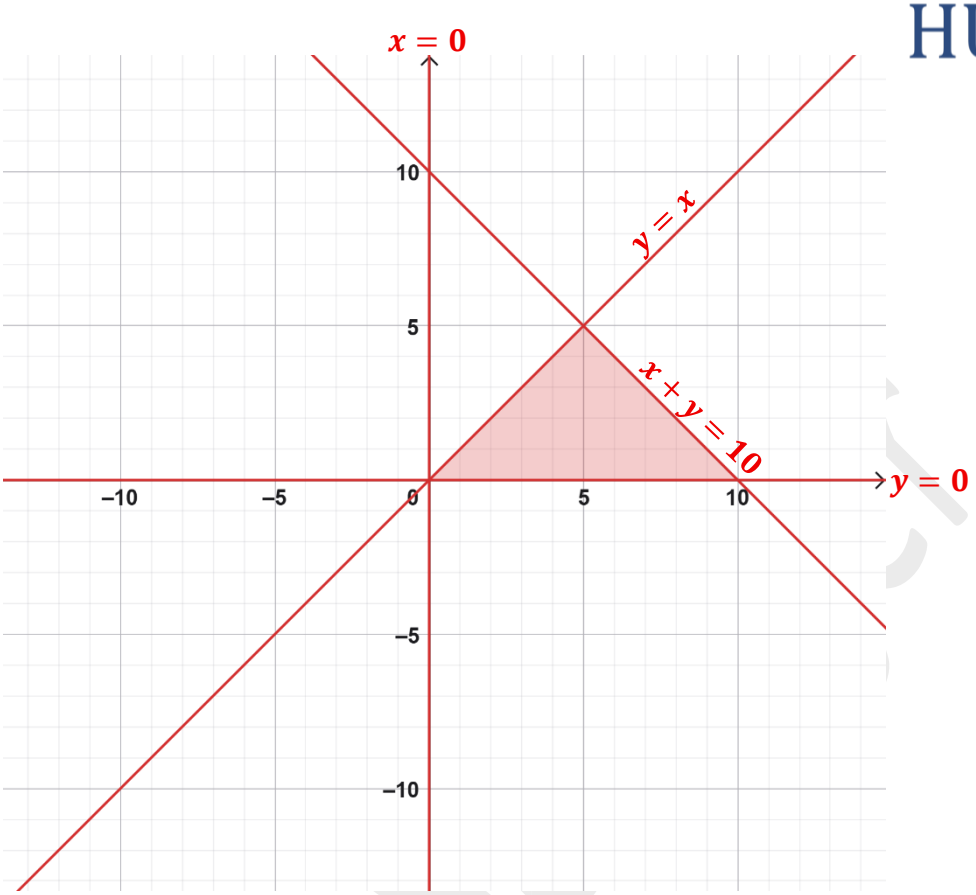
$x \geq 0$

$y \geq 0$

$x + y \leq 10$ and

$x \geq y$ [2]

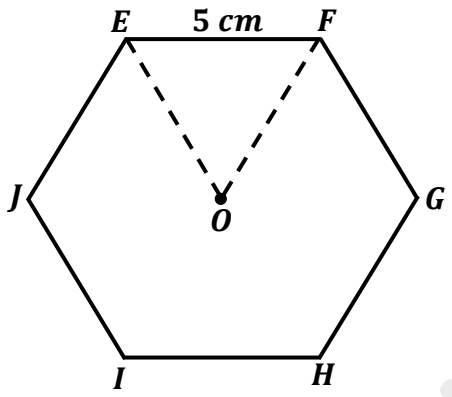




Total: 11 marks

Kerwin Sp

5. (a) The regular polygon $EFGHIJ$, shown below, has centre O. Triangle OEF is equilateral and $EF = 5$ cm.



(i) What is the name of the polygon shown above? [1]

The polygon has 6 sides.

Hence, it is a hexagon.

(ii) Calculate the perimeter of the polygon $EFGHIJ$. [1]

Perimeter of $EFGHIJ = 6 \times 5$

$= 30\ cm$

Kerwin Springer

(iii) Determine the size of each interior angle of the polygon. [2]

The sum of the interior angles of a polygon of n sides $= 180°(n-2)$.

The sum of the 6 interior angles of the given hexagon $= 180°(6-2)$

$= 180°(4)$

$= 720°$

Each interior angle of the hexagon $= \frac{720°}{6}$

$= 120°$

(iv) Show, by calculation, that the area of the polygon, to the nearest whole number, is 65 cm^2. [3]

Area of $\Delta OEF = \frac{1}{2}(OE)(OF)\sin E\hat{O}F$

$= \frac{1}{2}(5)(5)\sin 60°$

The hexagon comprises of 6 of these triangles.

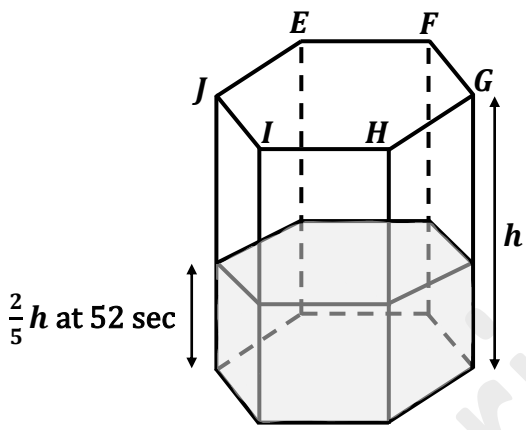
$\therefore$ Area of hexagon $= 6 \times \frac{1}{2}(5)(5)\sin 60°$

$= 64.95$

$= 65\ cm^2$ [to the nearest whole number]

Kerwin Springer

(b) A tank has a cross section with dimensions **identical** to the polygon $EFGHIJ$ in 5(a). Water is poured into the tank at a rate of 75 cm^3 per second. After 52 seconds the tank is $\frac{2}{5}$ full.



(i) Determine the capacity of the tank, in litres. [3]

Cross-sectional area of the tank = Area of the hexagon

$= 64.95\ cm^2$

After 52 seconds, the volume of water poured $= 75 \times 52$

$= 3900\ cm^3$

The tank is $\frac{2}{5}$ full.

Hence,

$\frac{2}{5}V = 3900$

$V = \frac{3900}{\frac{2}{5}}$

$V = 9750\ cm^3$

Kerwin Springer

Now,

$1000\ cm^3 = 1\ l$

$1\ cm^3 = \frac{1}{1000}$

$9750\ cm^3 = \frac{1}{1000} \times 9750$

$9750\ cm^3 = 9.75\ l$

∴ The capacity of the tank is 9.75 litres.

(ii) Calculate the height, h, in metres, of the tank. [2]

$V = \text{Cross-sectional area} \times h$

$V = 64.95h$

Now,

$\frac{2}{5}V = 3900$

$\frac{2}{5} \times 64.95 \times h = 3900$

$25.98 \times h = 3900$

$h = \frac{3900}{25.98}$

$h = 150\ cm$ [to the nearest whole number]

We need to convert 150 cm to metres, m.

$100\ cm = 1\ m$

$1\ cm = \frac{1}{100} m$

$150\ cm = \frac{1}{100} \times 150$

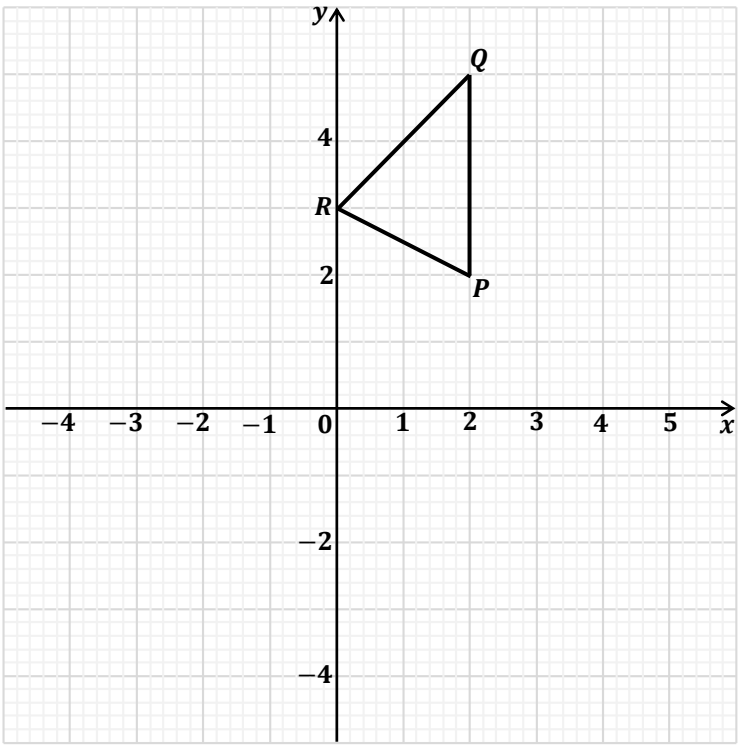
$150\ cm = 1.50\ m$ [to 2 decimal places]

$\therefore$ The height, h, of the tank is $1.50\ m$.

Total: 12 marks

Kerwin Springer

6. The diagram below shows triangle PQR.



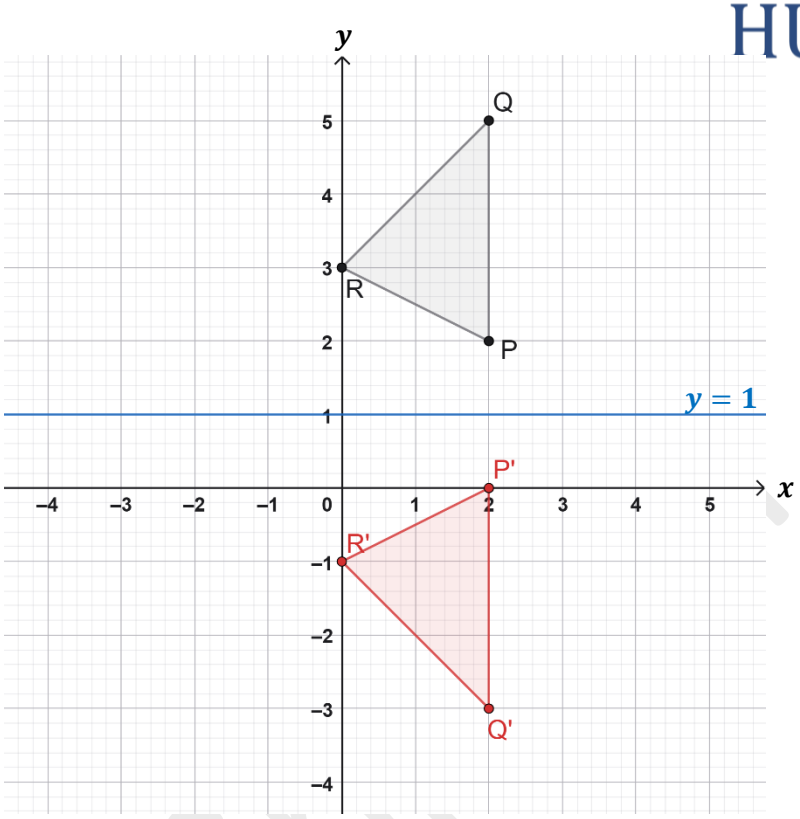
(a) State the coordinates of R. [1]

From the diagram, the coordinates of R are $(0, 3)$.

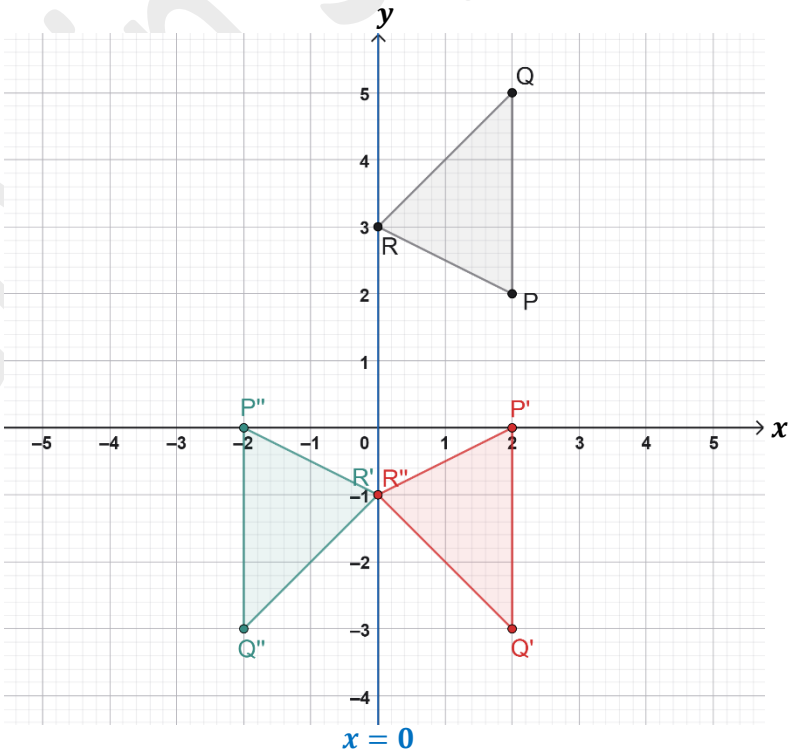
(b) On the diagram above, draw

(i) $\Delta P'Q'R'$, a reflection of ΔPQR in the line $y = 1$ [2]

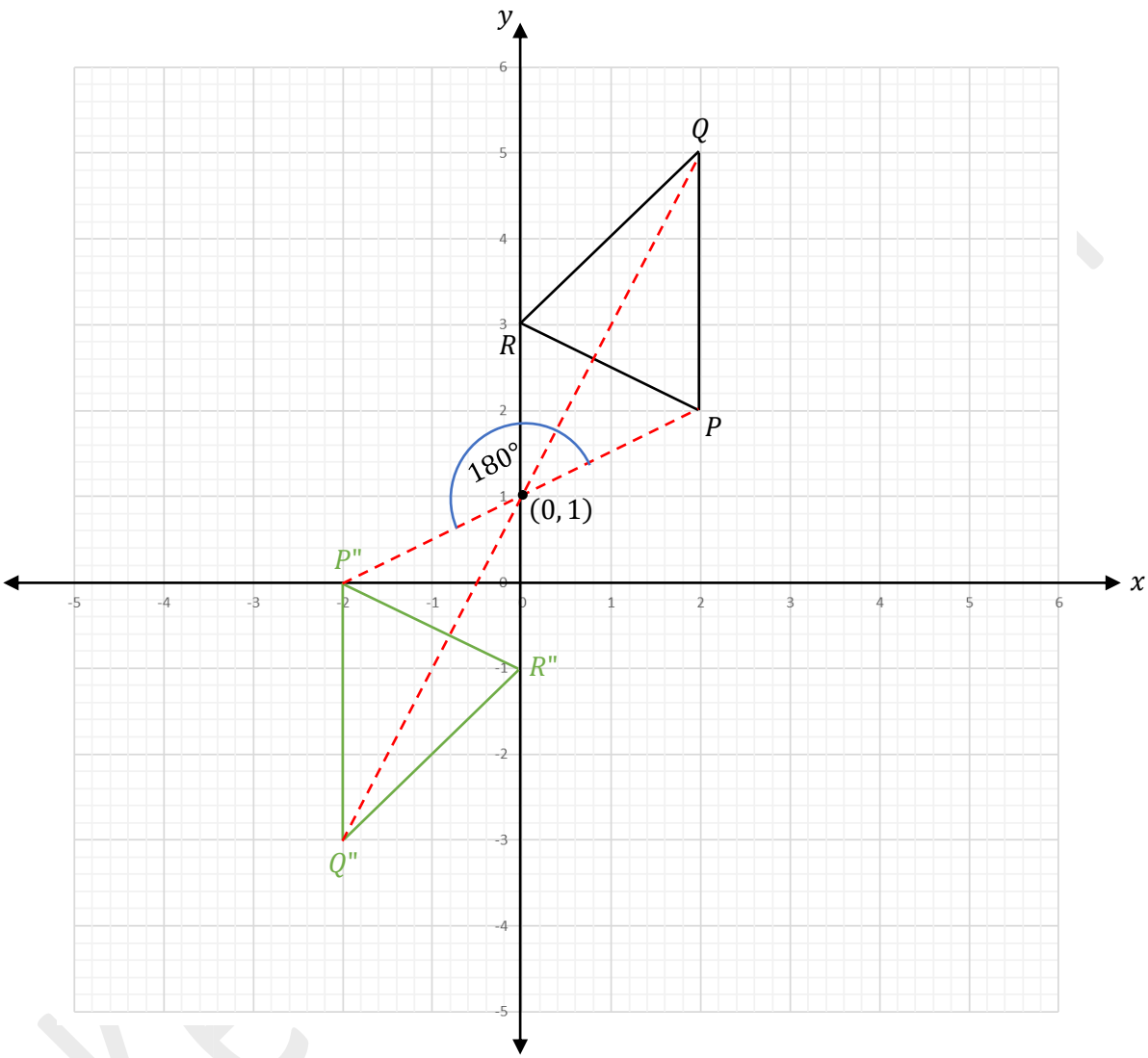
Kerwin Singer



(ii) $\Delta P''Q''R''$, a reflection of $\Delta P'Q'R'$ in the line $x = 0$ [2]



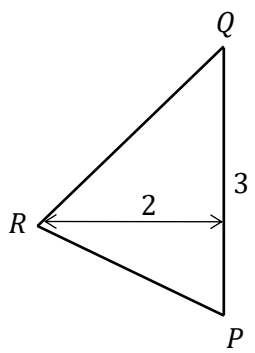
(c) Describe, fully, the single transformation that maps $\Delta P"Q"R"$ onto ΔPQR. [3]



The transformation that maps $\Delta P"Q"R"$ onto ΔPQR is a rotation of 180° about the point $(0, 1)$.

(d) Triangle PQR undergoes an enlargement of scale factor 2. Calculate the area of its image. [2]

Consider ΔPQR:



Area of $\Delta PQR = \frac{b \times h}{2}$

$= \frac{3 \times 2}{2}$

$= 3$ square units

Area of enlarged triangle $= 3 \times (scale\ factor)^2$

$= 3 \times (2)^2$

$= 3 \times 4$

$= 12$ square units

Total: 10 marks

Kerwin Springer

7. (a) The marks obtained by 10 students in a test, scored out of 60, are shown below.

29	38	26	42	38
45	35	37	38	31

For the data above, determine

(i) the range [1]

The highest mark is 45.

The lowest mark is 26.

Range = Highest mark − Lowest mark

$= 45 - 26$

$= 19$

(ii) the median [1]

Arranging the data in ascending order gives:

~~26~~ ~~29~~ ~~31~~ ~~35~~ 37 38 ~~38~~ ~~38~~ ~~42~~ ~~45~~

Median $= \frac{37+38}{2}$

$= 37.5$

Kerwin Springer

(iii) the interquartile range [2]

Arranging the data in ascending order gives:

26 29 31 35 37 38 38 38 42 45

The lower quartile, $Q_1 = 31$.

The upper quartile, $Q_3 = 38$.

Interquartile Range $= Q_3 - Q_1$

$= 38 - 31$

$= 7$

(iv) the probability that a student chosen at random scores less than half the total marks in the test. [2]

The total marks in the test $= 60$.

Half of the total marks $= \frac{60}{2}$

$= 30$

$= P(student\ scores\ less\ than\ half\ the\ total)$

$= \frac{Number\ of\ students\ who\ scored\ less\ than\ 30}{Total\ number\ of\ students}$

$= \frac{2}{10}$

$= \frac{1}{5}$ or 0.2 or 20%

Kerwin Springer

(b) The frequency distribution below shows the masses, in kg, of 50 adults prior to the start of a fitness programme.

Mass (kg)	Midpoint	Frequency
$60 - 64$	62	8
$65 - 69$	67	11
$70 - 74$	72	15
$75 - 79$	77	9
$80 - 84$	82	5
$85 - 89$	87	2

On the grid **on page 23**, using a scale of 2 cm to represent 5 units on the x-axis and 1 cm to represent 1 unit on the y-axis, draw a frequency polygon to represent the information in the table. [6]

Points to plot:

$(57, 0)$

$(62, 8)$

$(67, 11)$

$(72, 15)$

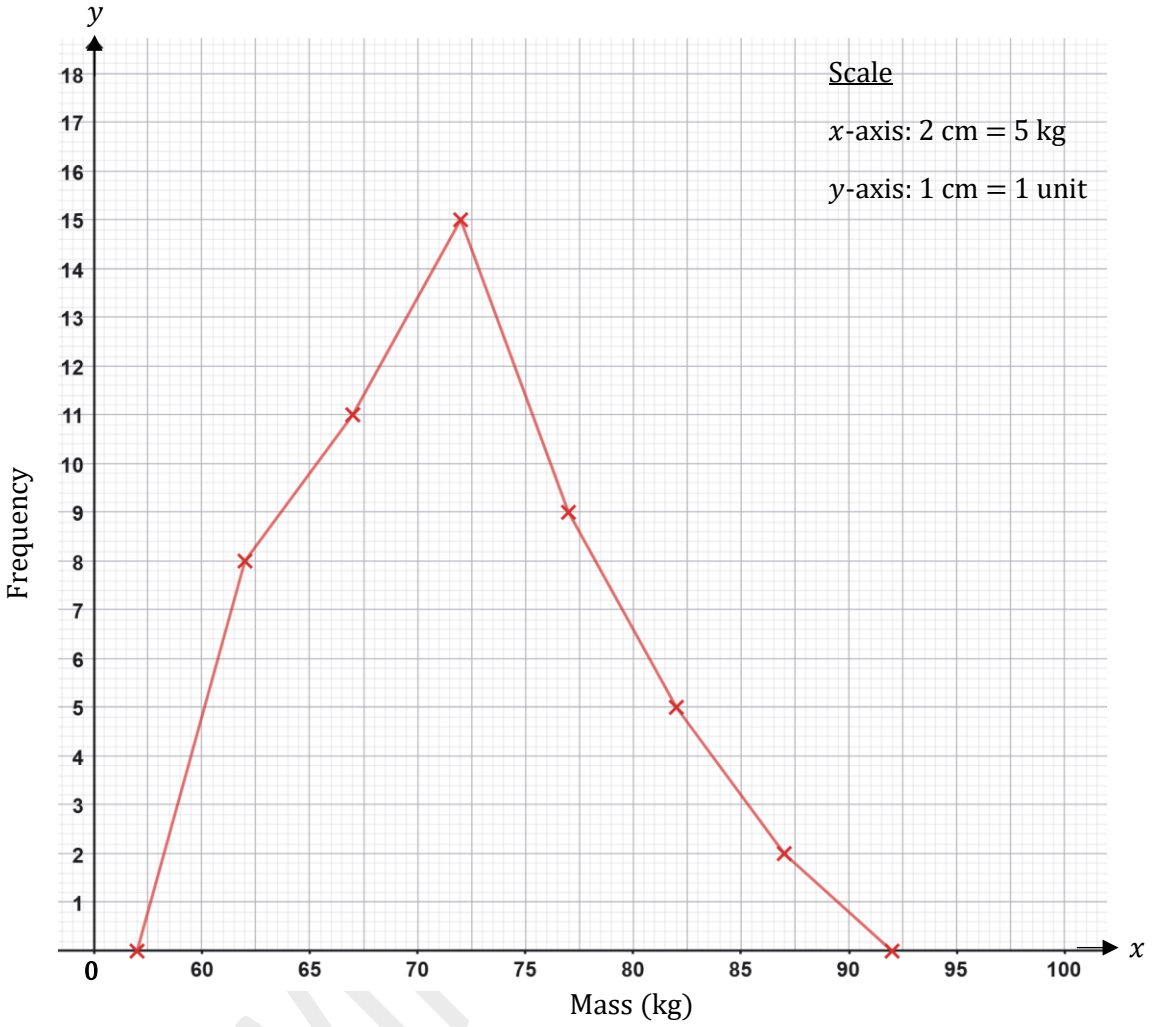
$(77, 9)$

$(82, 5)$

$(87, 2)$

$(92, 0)$

Kerwin Springer



Total: 12 marks

Best Online Lessons in the Caribbean
WhatsApp **+1868-310-1306** for more information

8. A sequence of figures is made from toothpicks of unit length. The first three figures in the ssequence are shown below.

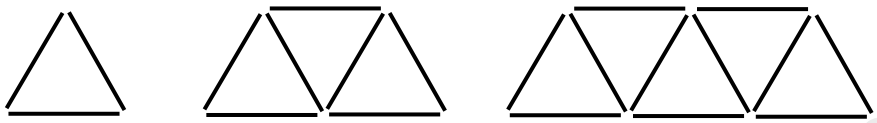


Figure 1 **Figure 2** **Figure 3**

(a) Draw Figure 4 of the sequence. [2]

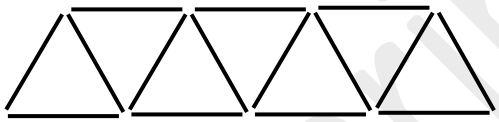


Figure 4

(b) Study the pattern of numbers in each row of the table below. Each row relates to one of the figures in the sequence of figures above. Some rows have not been included in the table.

Complete the rows numbered (i), (ii), (iii) and (iv).

Kerwin Springer

	Figure	Number of Toothpicks in Pattern	Perimeter of Figure	
	1	3	$0 + 1 + 2 = 3$	
	2	7	$1 + 2 + 2 = 5$	
	3	11	$2 + 3 + 2 = 7$	
(i)	4	15	$3 + 4 + 2 = 9$	[2]
	⋮			
(ii)	20	79	$19 + 20 + 2 = 41$	[2]
	⋮			
(iii)	32	127	$31 + 32 + 2 = 65$	[2]
	⋮			
(iv)	n	$4n - 1$	$(n - 1) + n + 2 = 2n + 1$	[2]

(i) For Figure 4, number of toothpicks $= 4(4) - 1$

$= 16 - 1$

$= 15$

(ii) For Figure 4, number of toothpicks $= 4(20) - 1$

$= 80 - 1$

$= 79$

Kerwin Springer

(iii) $4n - 1 = 127$

$4n = 127 + 1$

$4n = 128$

$n = \frac{128}{4}$

$n = 32$

(iv) For Figure n, number of toothpicks $= 4n - 1$

Total: 10 marks

Kerwin Springer

SECTION II

Answer TWO questions in this section.

ALGEBRA AND RELATIONS, FUNCTIONS AND GRAPHS

9. (a) (i) Show, by calculation, that the EXACT roots of the quadratic equation

$x^2 + 2x - 5 = 0$ are $-1 \pm \sqrt{6}$. [3]

$x^2 + 2x - 5 = 0$ is of the form $ax^2 + bx + c = 0$ where $a = 1, b = 2$ and $c = -5$.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-2 \pm \sqrt{(2)^2 - 4(1)(-5)}}{2(1)}$$

$$= \frac{-2 \pm \sqrt{4 + 20}}{2}$$

$$= \frac{-2 \pm \sqrt{24}}{2}$$

$$= \frac{-2 \pm \sqrt{4 \times 6}}{2}$$

$$= \frac{-2 \pm \sqrt{4}\sqrt{6}}{2}$$

$$= \frac{-2 \pm 2\sqrt{6}}{2}$$

Kerwin Springer

$$= \frac{2(-1 \pm \sqrt{6})}{2}$$

$$= -1 \pm \sqrt{6}$$

(ii) Hence, or otherwise, solve the simultaneous equations

$$2 + x = y$$

$$xy = 5 \qquad [4]$$

$2 + x = y$ ...Equation 1

$xy = 5$ ...Equation 2

Substituting Equation 1 into Equation 2 gives:

$$x(2 + x) = 5$$

$$2x + x^2 = 5$$

$$x^2 + 2x - 5 = 0$$

The exact roots of the quadratic equation $x^2 + 2x - 5 = 0$ are $-1 \pm \sqrt{6}$.

Hence, $x = -1 - \sqrt{6}$ and $x = -1 + \sqrt{6}$.

When $x = -1 - \sqrt{6}$,

$$y = 2 + (-1 - \sqrt{6})$$

$$y = 2 - 1 - \sqrt{6}$$

$$y = 1 - \sqrt{6}$$

Kerwin Springer

When $x = -1 + \sqrt{6}$,

$y = 2 + (-1 + \sqrt{6})$

$y = 2 - 1 + \sqrt{6}$

$y = 1 + \sqrt{6}$

$\therefore x = -1 - \sqrt{6}$ and $y = 1 - \sqrt{6}$ or $x = -1 + \sqrt{6}$ and $y = 1 + \sqrt{6}$.

(b) The incomplete table below shows values of x and y for the function $y = 2^x$ for integer values of x from -1 to 4.

x	-1	0	1	2	3	4
y		1			8	

(i) Complete the table for the function $y = 2^x$. [2]

When $x = -1$,

$y = 2^{-1}$

$y = \frac{1}{2}$

When $x = 1$,

$y = 2^1$

$y = 2$

When $x = 2$,

$y = 2^2$

$y = 4$

When $x = 4$,

$y = 2^4$

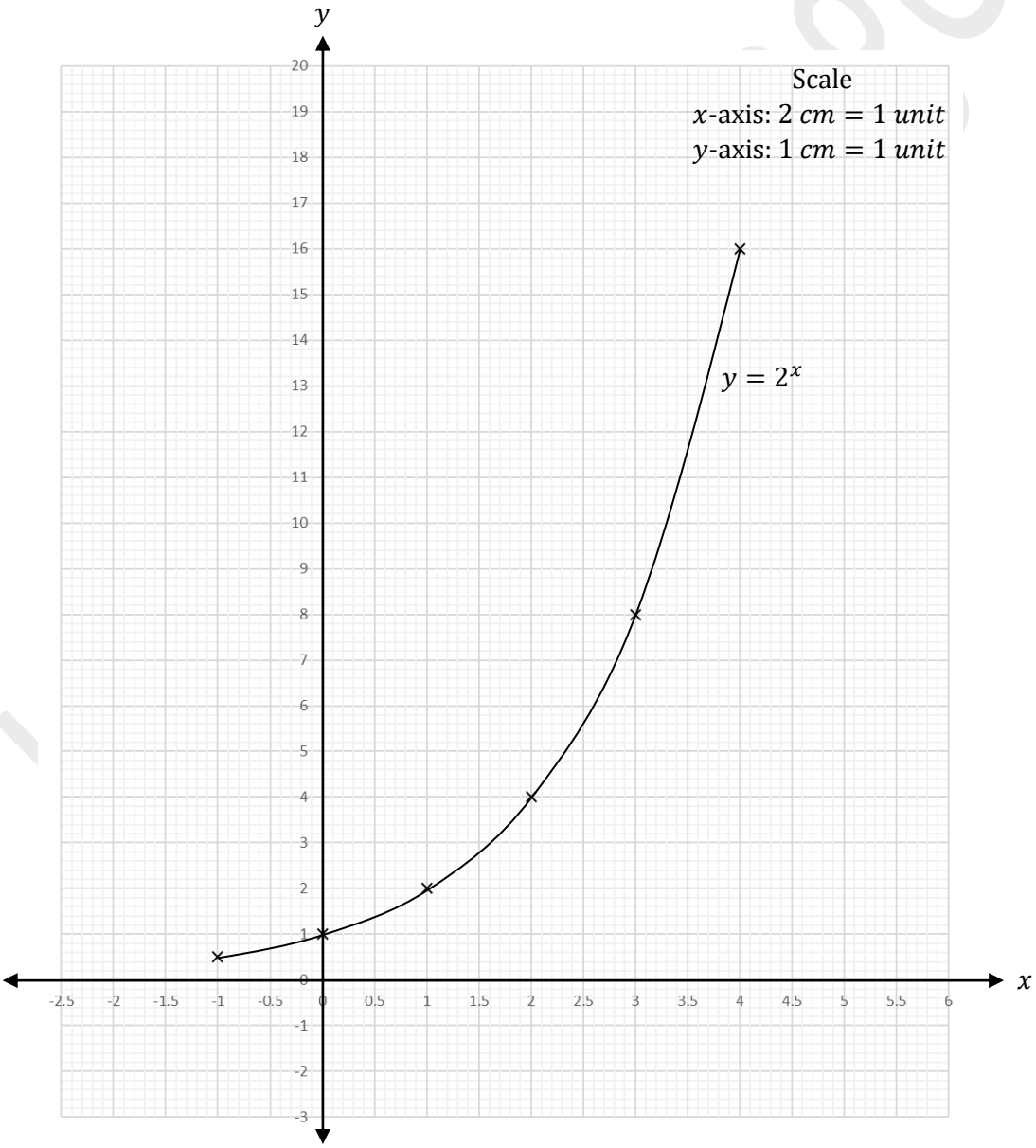
$y = 16$

Kerwin Springer

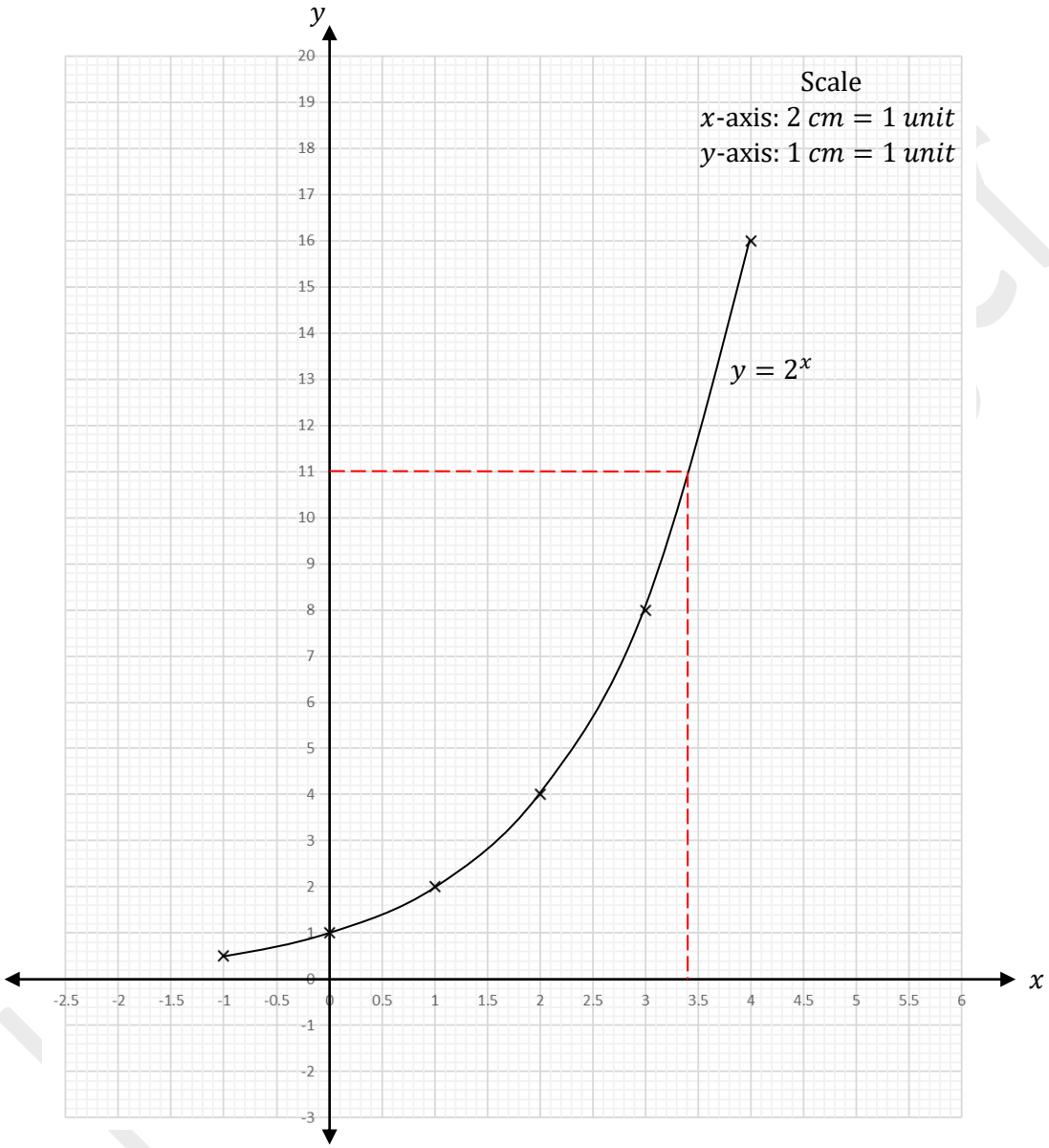
The completed table is shown below:

x	-1	0	1	2	3	4
y	$\frac{1}{2}$	1	2	4	8	16

(ii) On the grid provided **on page 27**, draw the graph of $y = 2^x$, using a scale of 2 cm to represent 1 unit on the x-axis and 1 cm to represent 1 unit on the y-axis. [4]



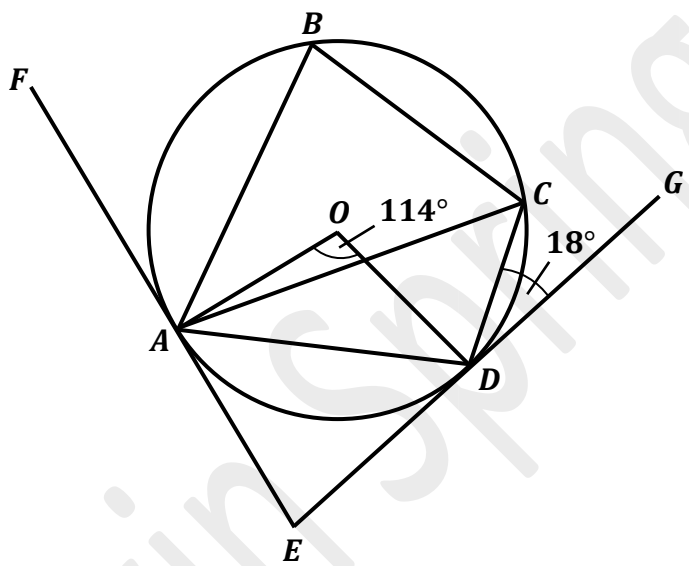
(iii) Drawing appropriate lines on your graph, determine the value of x for which $2^x = 11$. [2]



$\therefore$ When $2^x = 11$, $x = 3.4$.

Total: 15 marks

10. (a) The diagram below, **not drawn to scale**, shows a circle with centre O. The points A, B, C and D are on the circumference of the circle. EAF and EDG are tangents to the circle at A and D respectively. $A\hat{O}D = 114°$ and $C\hat{D}G = 18°$.



Calculate, giving reasons for EACH step of your answer, the measure of

(i) $A\hat{C}D$ [2]

The angle $A\hat{O}D$ subtended by a chord (AD) at the center of a circle, O, is twice the angle that the chord subtends at the circumference, standing on the same arc.

Hence,

$A\hat{C}D = \frac{1}{2}A\hat{O}D$

Kerwin Springer

$= \frac{1}{2}(114°)$

$= 57°$

(ii) $A\hat{E}D$ [2]

The angle between a tangent (EA and ED) to a circle and a radius (OA and OD) at the point of contact (A and D) is a right angle.

Hence, $O\hat{A}E = O\hat{D}E$

$= 90°$

Consider the quadrilateral $AODE$.

The sum of the interior angles of a quadrilateral is 360°.

$A\hat{E}D = 360° - (90° + 114° + 90°)$

$= 360° - 294°$

$= 66°$

(iii) $O\hat{A}C$ [2]

The angle between a tangent (DG) to a circle and a chord (DC) at the point of contact (D) is equal to the angle $(D\hat{A}C)$ in the alternate segment.

Since OA and OD are both radii of the same circle, then $OA = OD$ and ΔOAD is isosceles.

Kerwin Springer

$O\hat{A}D = O\hat{D}A$

$= \frac{180° - 114°}{2}$

$= \frac{66°}{2}$

$= 33°$

The base angles of an isosceles triangle are equal and the sum of the interior angles in a triangle is 180°.

Hence,

$O\hat{A}C = 33° - 18°$

$= 15°$

(iv) $A\hat{B}C$ [2]

The sum of the interior angles in a triangle add up to 180°.

$A\hat{D}C = 180° - (18° + 57°)$

$= 180° - 75°$

$= 105°$

The opposite angles of a cyclic quadrilateral are supplementary.

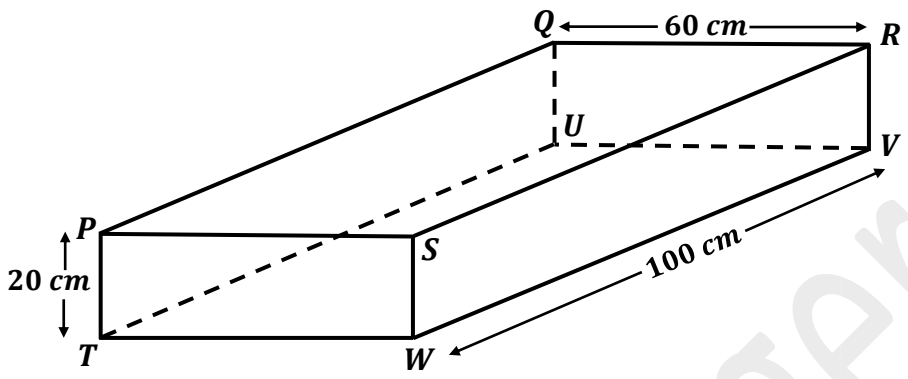
Hence,

$A\hat{B}C = 180° - 105°$

$= 75°$

Kerwin Springer

(b) The diagram below shows a cuboid.

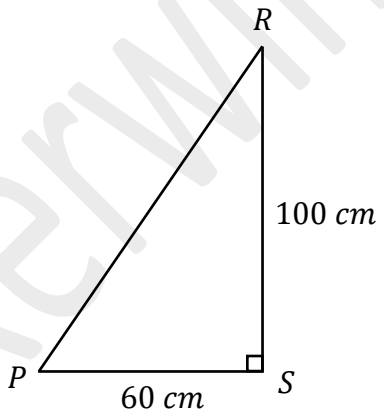


Give your answer correct to one decimal place.

(i) A straight adjustable wire connects R to P along the top of the cuboid.

Calculate the length of the wire RP. [1]

Consider ΔPRS:



Using Pythagoras' Theorem:

$RP^2 = PS^2 + SR^2$

$RP^2 = (60)^2 + (100)^2$

Kerwin Springer

$RP^2 = 3600 + 10\ 000$

$RP^2 = 13\ 600$

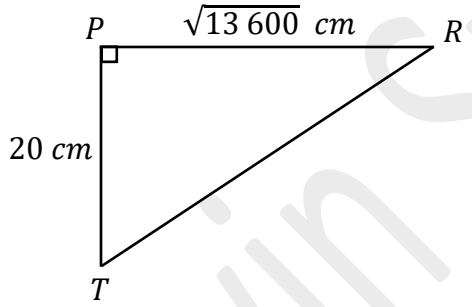
$RP = \sqrt{13\ 600}$

$RP = 116.6\ cm$ [correct to 1 decimal place]

(ii) The connection at P is now adjusted and moved to T.

Calculate the length of the wire RT. [2]

Consider ΔPRT:



Using Pythagoras' Theorem:

$RT^2 = PT^2 + PR^2$

$RT^2 = (20)^2 + \left(\sqrt{13\ 600}\right)^2$

$RT^2 = 400 + 13\ 600$

$RT^2 = 14\ 000$

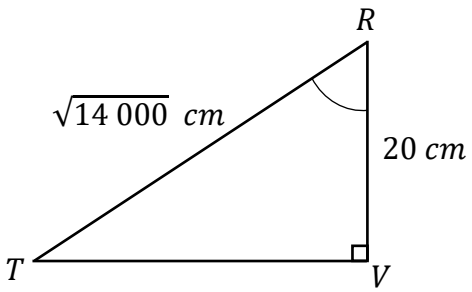
$RT = \sqrt{14\ 000}$

$RT = 118.3\ cm$ [correct to 1 decimal place]

Kerwin Springer

(iii) Calculate the angle TRV. [2]

Consider ΔTRV:



$\cos T\hat{R}V = \frac{VR}{RT}$

$\cos T\hat{R}V = \frac{20}{\sqrt{14\ 000}}$

$\cos T\hat{R}V = 0.169$

$T\hat{R}V = \cos^{-1}(0.169)$

$T\hat{R}V = 80.3°$ [correct to 1 decimal place]

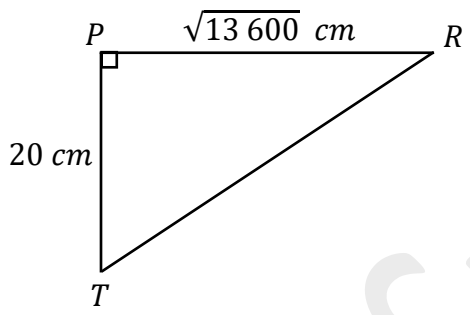
Kerwin Springer

(iv) Complete the following statements: [2]

The size of the angle through which the wire moves from RP to RT is ..... °.

An angle which is the same in size as RTV is .....

Consider ΔPRT:



From RP to RT, the wire moves through $P\hat{R}T$.

$\tan P\hat{R}T = \frac{PT}{PR}$

$\tan P\hat{R}T = \frac{20}{\sqrt{13\,600}}$

$\tan P\hat{R}T = 0.171$

$P\hat{R}T = \tan^{-1}(0.171)$

$P\hat{R}T = 9.7°$ (to 1 decimal place)

$R\hat{T}V$ has the same size as $P\hat{T}V$ (or $S\hat{U}W$ or $Q\hat{W}U$).

The completed statements are as follows:

Kerwin Springer

The size of the angle through which the wire moves from RP to RT is

9.7°

An angle which is the same in size as angle RTV is angle PRT

Total: 15 marks

Kerwin Springer

11. (a) Given the vectors $\overrightarrow{OP} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$, $\overrightarrow{PQ} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ and $\overrightarrow{RS} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$,

(i) determine the vector $\overrightarrow{OQ}$ [2]

$\overrightarrow{OP} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

$\overrightarrow{PQ} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$

So, we have,

$\overrightarrow{OQ} = \overrightarrow{OP} + \overrightarrow{PQ}$

$= \begin{pmatrix} 3 \\ 4 \end{pmatrix} + \begin{pmatrix} -1 \\ 2 \end{pmatrix}$

$= \begin{pmatrix} 3 + (-1) \\ 4 + 2 \end{pmatrix}$

$= \begin{pmatrix} 2 \\ 6 \end{pmatrix}$

(ii) show that $\overrightarrow{OQ}$ is parallel to $\overrightarrow{RS}$, giving a reason for your answer. [1]

$\overrightarrow{RS} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

$= \frac{1}{2}\begin{pmatrix} 2 \\ 6 \end{pmatrix}$

$= \frac{1}{2}\overrightarrow{OQ}$

Since $\overrightarrow{RS}$ is a scalar multiple of $\overrightarrow{OQ}$, then $\overrightarrow{OQ}$ and $\overrightarrow{RS}$ are parallel.

Kerwin Springer

(b) XYZ is a triangle and M is the midpoint of XZ.

$\overrightarrow{XY} = \boldsymbol{a}$ and $\overrightarrow{YZ} = \boldsymbol{b}$.

Express the following vectors in terms of $\boldsymbol{a}$ and $\boldsymbol{b}$, simplifying your answers where possible:

(i) $\overrightarrow{XZ}$ [1]

$\overrightarrow{XZ} = \overrightarrow{XY} + \overrightarrow{YZ}$

$\overrightarrow{XZ} = \boldsymbol{a} + \boldsymbol{b}$

(ii) $\overrightarrow{MY}$ [3]

M is the midpoint of XZ.

$\overrightarrow{XM} = \frac{1}{2}\overrightarrow{XZ}$

$= \frac{1}{2}(\boldsymbol{a} + \boldsymbol{b})$

$\therefore \overrightarrow{MX} = -\frac{1}{2}(\boldsymbol{a} + \boldsymbol{b})$

$\overrightarrow{MY} = \overrightarrow{MX} + \overrightarrow{XY}$

$= -\frac{1}{2}(\boldsymbol{a} + \boldsymbol{b}) + \boldsymbol{a}$

$= -\frac{1}{2}\boldsymbol{a} - \frac{1}{2}\boldsymbol{b} + \boldsymbol{a}$

$= \frac{1}{2}\boldsymbol{a} - \frac{1}{2}\boldsymbol{b}$

$\therefore \overrightarrow{MY} = \frac{1}{2}(\boldsymbol{a} - \boldsymbol{b})$

Kerwin Springer

(c) The matrices A and B are given as $A = \begin{pmatrix} -1 & 0 \\ 3 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} -3 & 2 \\ 1 & 1 \\ 4 & 6 \end{pmatrix}$.

(i) Determine A^{-1}, the inverse of A. [2]

$$A = \begin{pmatrix} -1 & 0 \\ 3 & 2 \end{pmatrix}$$

$$\det A = ad - bc$$

$$= (-1)(2) - (0)(3)$$

$$= -2 - 0$$

$$= -2$$

$$adj(A) = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 0 \\ -3 & -1 \end{pmatrix}$$

$$\therefore A^{-1} = \frac{1}{\det A} \times adj(A)$$

$$= \frac{1}{-2}\begin{pmatrix} 2 & 0 \\ -3 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{2}{-2} & \frac{0}{-2} \\ \frac{-3}{-2} & \frac{-1}{-2} \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 0 \\ \frac{3}{2} & \frac{1}{2} \end{pmatrix}$$

Kerwin Springer

(ii) Show that $A^{-1}A = I$, the identity matrix. [2]

The identity matrix is $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

Now,

$$A^{-1} = \frac{1}{-2}\begin{pmatrix} 2 & 0 \\ -3 & -1 \end{pmatrix}$$

So, we have,

$$A^{-1}A = \frac{1}{-2}\begin{pmatrix} 2 & 0 \\ -3 & -1 \end{pmatrix}\begin{pmatrix} -1 & 0 \\ 3 & 2 \end{pmatrix}$$

$$= \frac{1}{-2}\begin{pmatrix} (2 \times -1) + (0 \times 3) & (2 \times 0) + (0 \times 2) \\ (-3 \times -1) + (-1 \times 3) & (-3 \times 0) + (-1 \times 2) \end{pmatrix}$$

$$= \frac{1}{-2}\begin{pmatrix} -2 + 0 & 0 + 0 \\ 3 + (-3) & 0 + (-2) \end{pmatrix}$$

$$= \frac{1}{-2}\begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{-2}{-2} & \frac{0}{-2} \\ \frac{0}{-2} & \frac{-2}{-2} \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Hence, $A^{-1}A = I$.

Kerwin Springer

(iii) Determine the matrix A^2. [2]

$$A^2 = A \times A$$

$$= \begin{pmatrix} -1 & 0 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 3 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} (-1 \times -1) + (0 \times 3) & (-1 \times 0) + (0 \times 2) \\ (3 \times -1) + (2 \times 3) & (3 \times 0) + (2 \times 2) \end{pmatrix}$$

$$= \begin{pmatrix} 1 + 0 & 0 + 0 \\ -3 + 6 & 0 + 4 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 3 & 4 \end{pmatrix}$$

$$\therefore A^2 = \begin{pmatrix} 1 & 0 \\ 3 & 4 \end{pmatrix}$$

(iv) (a) Explain why the matrix product AB is NOT possible. [1]

The number of columns of A which is 2 is not equal to the number of rows of B which is 3. Hence, the product AB is not possible.

(b) Without calculating, state the order of the matrix product BA. [1]

The size of the matrix B is 3×2.

The size of the matrix A is 2×2.

Kerwin Springer

Therefore, the size of the matrix product BA is 3×2.

Total: 15 marks

Kerwin Springer

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

Best Online Lessons in the Caribbean
WhatsApp **+1868-310-1306** for more information